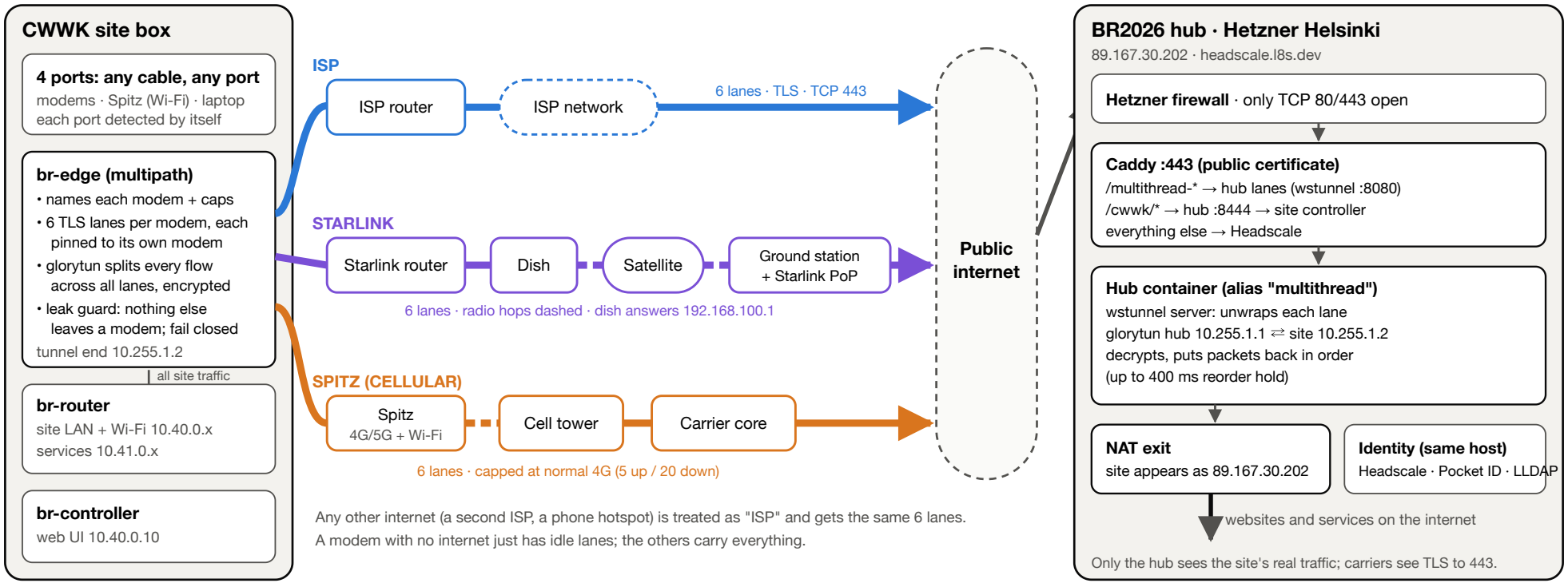
Black Relay: how the site reaches the internet

every modem carries its own encrypted lanes, over its own carrier, to the hub in Hetzner

November 2026



Remote access to the site controller (from anywhere)

Browser anywhere → `headscale.l8s.dev/cwwk/` Caddy on BR2026 → hub :8444 forwarded through the tunnel → controller 10.40.0.10 accepts the hub only → Pocket ID sign-in (id.l8s.dev) survivability = admin · headscale-users = view

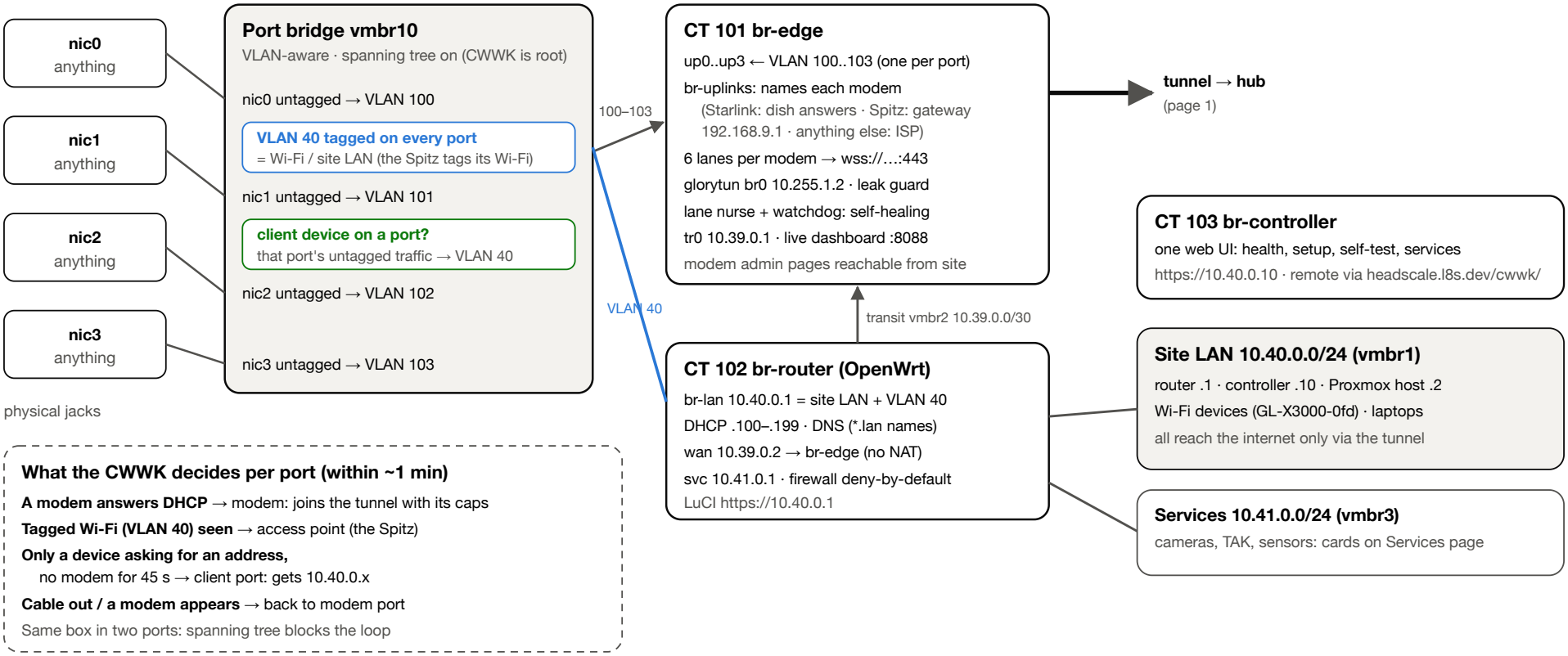
Needs the tunnel up. On site, the controller is also at https://10.40.0.10 (admin password = break-glass when the tunnel or Pocket ID is down).
The CWWK's own Tailscale (100.64.0.11) also rides the tunnel, so a tunnel outage takes it off the tailnet too.

ISP lanes · Starlink lanes · Spitz lanes · radio hop / public internet

Speed weights (up/down Mbit): ISP 100/150 · Starlink 10/150 · Spitz 5/20; downloads split evenly ISP/Starlink.

Inside the CWWK: any cable, any port

one port bridge, three containers, the site networks



What the CWWK decides per port (within ~1 min)

A modem answers DHCP → modem: joins the tunnel with its caps

Tagged Wi-Fi (VLAN 40) seen → access point (the Spitz)

Only a device asking for an address,
no modem for 45 s → client port: gets 10.40.0.x

Cable out / a modem appears → back to modem port

Same box in two ports: spanning tree blocks the loop

Reference addresses, web interfaces, security controls, tuning

Networks and addresses

Network	Range	Where
Site LAN + Wi-Fi	`10.40.0.0/24`	router .1 · controller .10 · host .2 · DHCP .100–.199 · VLAN 40 on every port
Services	`10.41.0.0/24`	router .1 · cameras, TAK, sensors
Router ⇄ multipath	`10.39.0.0/30`	br-edge .1 · br-router .2 (internal)
Tunnel overlay	`10.255.1.0/30`	hub .1 · site .2
Port VLANs	`100–103`	port i → br-edge up i (internal to the CWWK)
Modem networks	whatever each modem hands out	e.g. Starlink 192.168.1.x · Spitz 192.168.9.x · ISP 192.168.8.x
Tailnet	`100.64.0.0/10`	CWWK host 100.64.0.11 (rides the tunnel)

Containers on the CWWK

ID	Name	Role
101	br-edge	modem detection, lanes, bonding, tunnel, leak guard, live dashboard
102	br-router	OpenWrt: DHCP, DNS, firewall, LuCI
103	br-controller	single web UI for the whole box

Web interfaces

What	On site	Anywhere
Controller	`https://10.40.0.10`	`https://headscale.l8s.dev/cwwk/`
Status (no login)	`https://status.lan`	–
Router (LuCI)	`https://10.40.0.1`	–
Proxmox	`https://10.40.0.2:8006`	`https://100.64.0.11:8006` `(tailnet)`
Modem admin pages	controller → Setup → Quick links	–

Logins are held by the site operator, never in this document.

Security controls

- **Fail closed.** Site traffic can only leave through the tunnel; if it is down, traffic is dropped, never sent out a raw modem.
- **Leak guard.** Only the encrypted lanes to the hub (TLS 443, each pinned to its modem), DHCP and modem admin pages may use a modem.
- **Hostile modems can't steer traffic.** Routes a modem pushes are ignored, leases overlapping the site are refused, and site addresses always route via the router.
- **Hub inside existing ports.** No new public ports on BR2026: lanes ride HTTPS 443 behind Caddy; a secret path plus the tunnel's own keys authenticate the site.
- **Hub egress is public-only.** Site traffic can't reach the hub's internal networks or private ranges; the hub's own address only on 80/443.
- **No rogue DHCP on any port.** DHCP replies and IPv6 router adverts from a port only reach the modem side; nothing plugged in can take over the site LAN's addressing.
- **Remote control needs Pocket ID.** The controller's remote listener accepts only the hub, and every person signs in (admin or view-only by group).
- **Least privilege.** Proxmox token: view + power only; multipath changes through fixed commands; the three core containers can't be stopped from the UI.
- **No side doors.** SSH is key-only; no tailnet forwarders; the on-site password works on the site LAN only (break-glass).
- **Self-healing.** Moved cables, DHCP changes, stuck lanes and a wedged tunnel recover on their own; the CWWK is back about 2 min after power-on.

Tuning (November 2026)

Setting	Value
Mixing	per-packet random, weighted by each modem's speed caps
Speed caps (up/down Mbit)	ISP 100/150 · Starlink 10/150 · Spitz 5/20 · follow the modem, not the port
Lanes	6 HTTPS connections per modem
Reorder hold	400 ms
Tunnel MTU	1400 (fixed for HTTPS lanes)
Round trip to hub	~110–130 ms

Starlink's upload is ~9 Mbit raw, so uploads lean on the ISP by design. Change caps on the controller's Multipath page.